

EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

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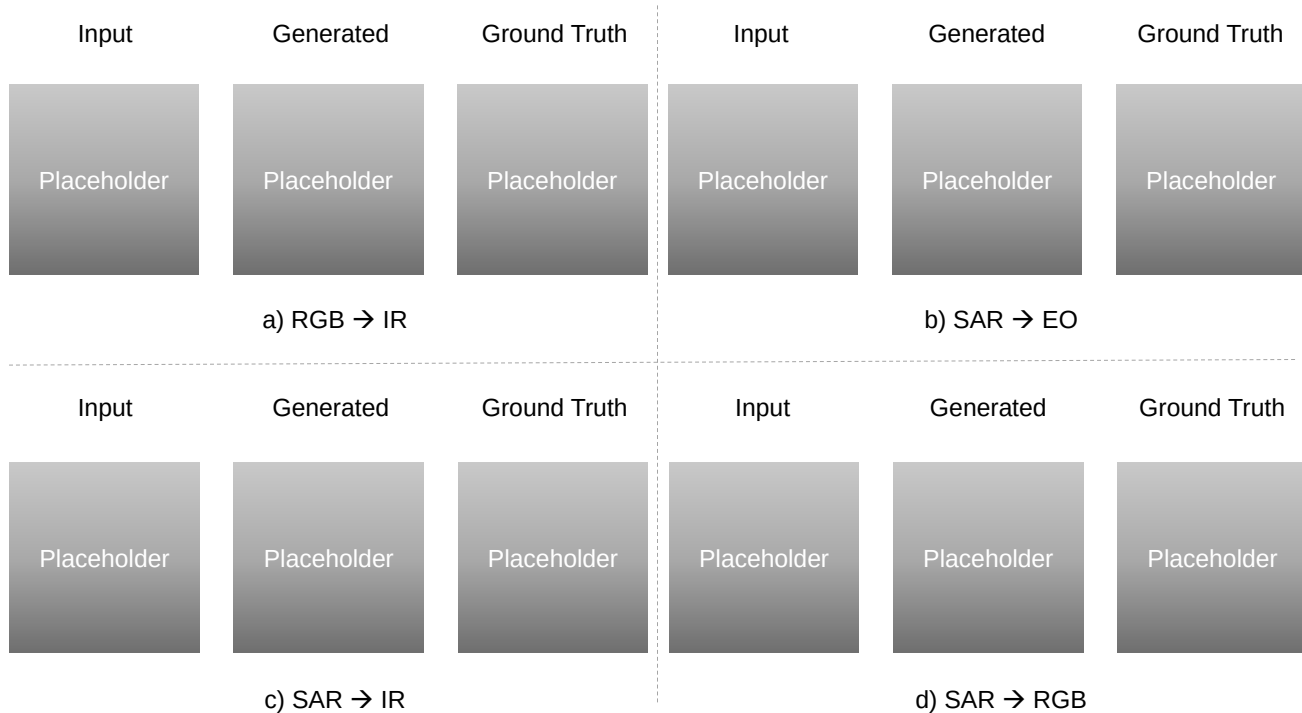


Figure 1. Teaser: The proposed EarthBridge framework for multi-modal aerial-view image translation across Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR) modalities. Our method accurately bridges the domain gap, preserving structural integrity while synthesizing high-fidelity target textures.

Abstract

001 Cross-modal image-to-image translation among Electro-
 002 Optical (EO), Infrared (IR), and Synthetic Aperture Radar
 003 (SAR) sensors is essential for comprehensive multi-modal
 004 aerial-view analysis. However, translating between these
 005 modalities is notoriously difficult due to their distinct elec-
 006 tromagnetic signatures and geometric characteristics. This
 007 paper presents **EarthBridge**, a high-fidelity translation
 008 framework developed for the 4th Multi-modal Aerial View
 009 Image Challenge – Translation (MAVIC-T). EarthBridge

is founded on **Diffusion Bridge Implicit Models (DBIM)**,
 which we generalize using non-Markovian bridge pro-
 cesses to enable efficient, high-quality deterministic sam-
 pling in as few as **[TODO: 40]** steps. Our method em-
 ploys a channel-concatenated UNet denoiser trained with
 Karras-weighted bridge scalings and a specialized "boot-
 ing noise" initialization to handle the inherent ambiguity
 in cross-modal mappings. To accommodate high-resolution
 1024×1024 imagery, we implement a multi-stage training
 protocol involving random-crop pre-training followed by
 full-resolution fine-tuning. We evaluate EarthBridge across

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all four challenge tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), achieving superior spatial detail and spectral accuracy. Our solution achieved a composite score of [TODO: 0.XXX], securing a [TODO: Rank] position on the MAVIC-T leaderboard.

1. Introduction

The Multi-modal Aerial View Image Challenge for cross-domain image translation [1, 19, 20] is a leading challenge held in conjunction with the Perception Beyond the Visible Spectrum (PBVS) workshop from 2023 to 2025. Data collected across multiple sensing modalities—such as Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR)—provide complementary perspectives on the same scene. However, data coverage limitations and modality-specific constraints often hinder the full utilization of these diverse sources. For instance, while SAR sensors offer all-weather capabilities and can penetrate atmospheric obstructions, their imagery is complex to interpret, and publicly available datasets are scarce. Conversely, widely available EO imagery is limited by lighting conditions and atmospheric visibility. Domain translation seeks to bridge this gap by converting one modality into another, enabling cross-modal data synthesis and enhancing sensor interoperability. By focusing on translation tasks such as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the MAVIC-T challenge aims to bolster existing data volume, increase data diversity across regions, and serve as a platform for advancing conditioned image synthesis techniques, ultimately fostering the development of more adaptable and robust models for multi-modal image analysis.

SAR images possess unique attributes that present challenges for both human observers and vision artificial intelligence (AI) models to interpret, owing to their electromagnetic (EM) characteristics [10–13, 21].

In this paper, we propose **EarthBridge**, a diffusion-based framework for cross-modal aerial image translation built on Diffusion Bridge Implicit Models (DBIM) [44]. Our approach generalizes the diffusion bridge framework with non-Markovian bridge processes, enabling efficient deterministic sampling. By conditioning the bridge process on the source image via channel concatenation and training a UNet denoiser under Karras-weighted bridge scalings [15], EarthBridge achieves high-fidelity translation across disparate modalities. We utilize pixel-level modeling throughout to preserve the full spectral and spatial resolution of the remote sensing data.

Our findings indicate that the deterministic DBIM sampling significantly improves inference speed, achieving high-quality results in as few as [TODO: 40] sampling steps. Qualitatively, EarthBridge demonstrates a superior ability to reconstruct complex urban layouts and fine-

grained textures in challenging tasks such as SAR-to-RGB, effectively bridging the domain gap. Quantitatively, our solution achieves a composite task score of [TODO: 0.XXX] for the SAR→EO task and [TODO: 0.XXX] overall, ultimately placing [TODO: Rank] in the MAVIC-T challenge leaderboard.

Our main contributions are summarized as follows:

- We propose **EarthBridge**, a novel framework for paired cross-modal aerial image translation based on Diffusion Bridge Implicit Models (DBIM), which generalizes the bridge process with non-Markovian dynamics and enables efficient deterministic sampling.
- We develop a specialized training protocol using Karras-weighted bridge scalings and pixel-level modeling, optimized for the high dynamic range and resolution requirements of multimodal remote sensing data.
- We introduce a multi-stage training strategy involving random crop pre-training and full-resolution fine-tuning to effectively handle 1024×1024 translation tasks within practical memory constraints.
- We conduct comprehensive experiments on all four MAVIC-T tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), reporting state-of-the-art results and providing detailed quantitative and qualitative analysis.

2. Related Work

Translating across remote sensing (RS) modalities, such as SAR to optical (S2O), is critical for multisource data fusion. In this section, we provide a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing.

General Image-to-Image Translation. Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [14] and CycleGAN for unpaired translation [46]. Subsequent advances include contrastive learning for unpaired translation [23], zero-shot [24] and one-step translation with text-to-image models [25].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [45] define a variance-preserving bridge between data distributions. Related formulations include Brownian Bridge Diffusion Models (BBDM) [16] and its bidirectional extension BiBBDM [41], Image-to-Image Schrödinger Bridge (I^2SB) [17], Diffusion Schrödinger Bridge matching [34], Dual Diffusion Implicit Bridges [35],

and Direct Diffusion Bridge with data consistency [2]. Recent extensions include Consistency Diffusion Bridge [6], Diffusion Bridge Implicit Models [44], deterministic translation via denoising Brownian bridge with dual approximators [40], and adaptive domain shift for cross-modality translation [39].

Cross-Modality Translation. Image translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been addressed with probabilistic color correction [22], NIR imagery colorization [36], deep colorization [21, 42].

GAN-based RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). For instance, Wang et al. [38] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [9] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors.

Diffusion Models for RS Translation. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation tasks. Qin et al. [28] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [29] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Furthermore, Wang et al. [39] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

3. Preliminary

Problem Formulation Let $\mathcal{X} \subset \mathbb{R}^{C_s \times H \times W}$ denote the source domain and $\mathcal{Y} \subset \mathbb{R}^{C_t \times H \times W}$ denote the target domain, where C_s and C_t are the number of channels in the source and target modalities, respectively. Given a set of spatially-aligned training pairs $\{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the goal of cross-modal image-to-image translation is to learn a mapping $G : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes a composite distance between the generated image $\hat{y} = G(x)$ and the ground-truth target y . In the context

of the MAVIC-T challenge, four translation tasks are defined: SAR→EO ($1 \rightarrow 1$ channel, 256×256), SAR→RGB ($1 \rightarrow 3$, 1024×1024), SAR→IR ($1 \rightarrow 1$, 1024×1024), and RGB→IR ($3 \rightarrow 1$, 1024×1024).

Table 1. Task specifications for the MAVIC-T challenge.

Task	Modality Mapping	Resolution	Channels
SAR→EO	SAR → Electro-Optical	256^2	$1 \rightarrow 1$
SAR→RGB	SAR → RGB	1024^2	$1 \rightarrow 3$
SAR→IR	SAR → Infrared	1024^2	$1 \rightarrow 1$
RGB→IR	RGB → Infrared	1024^2	$3 \rightarrow 1$

Evaluation Metric The MAVIC-T challenge evaluates submissions using a composite score derived from three complementary metrics: LPIPS [43] (Learned Perceptual Image Patch Similarity, based on VGG-16) for perceptual fidelity, FID [8] (Fréchet Inception Distance, based on InceptionV3) for distributional similarity, and L1 norm for pixel-level structural accuracy. These are combined into a task score:

$$S_{\text{task}} = \frac{\frac{2}{\pi} \arctan(\text{FID}) + \text{LPIPS} + \text{L1}}{3}, \quad (1)$$

where the arctan normalization maps FID into the $[0, 1]$ range to balance its contribution against the other two metrics. The overall score is the average of the four task scores, with a penalty of +1 added for each unattempted domain, encouraging participants to develop generalizable solutions across all modality pairs.

4. Diffusion Bridge Foundation

Diffusion models have recently revolutionized generative modeling, yet they are fundamentally designed to transport complex distributions to a standard Gaussian prior. Adapting them to translate between two arbitrary distributions, such as heterogeneous remote sensing modalities, requires conditioning the diffusion process on a fixed endpoint. This is achieved via Doob’s h -transform [5, 30].

4.1. Stochastic Bridges via h -transform

Given a base diffusion process $dx_t = f(x_t, t)dt + g(t)dw_t$, the conditioned SDE that almost surely reaches a prescribed target $x_0 = y$ at $t = 0$ starting from a source $x_T = x$ at $t = T$ is given by:

$$dx_t = f(x_t, t)dt + g(t)dw_t + g(t)^2 \nabla_{x_t} \log p(x_0 = y | x_t)dt, \quad (2)$$

where $p(x_0 = y | x_t)$ is the transition density of the original process. When the endpoints are fixed, the resulting conditioned process is called a *diffusion bridge* [3, 7, 18,

27, 32, 33], which forms the foundational mechanism for transporting one state to another. In the context of cross-modal translation, x represents the source modality (e.g., SAR) and y the target modality (e.g., optical). This formulation ensures that the generative trajectory is directly constrained by both endpoints, providing a more principled framework for paired translation than standard conditional diffusion.

4.2. Denoising Diffusion Bridge Models

Denoising Diffusion Bridge Models (DDBM) [45] provide a tractable realization of Eq. 2. Under a variance-preserving (VP) schedule, the forward bridge process constructs intermediate states z_t as:

$$z_t = a_t x + b_t y + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

where a_t, b_t, σ_t are schedule-dependent coefficients. While DDBM excels in high-fidelity generation, its inference typically relies on simulating Eq. 2 via computationally expensive numerical solvers (e.g., Heun or stochastic samplers), often requiring hundreds of function evaluations (NFE) for optimal results. In this work, we adopt the more efficient *implicit* formulation to accelerate this process.

5. EarthBridge: Diffusion Bridge Implicit Models for RS Translation

We present our EarthBridge approach based on **Diffusion Bridge Implicit Models (DBIM)** [44], which generalizes the diffusion bridge framework with non-Markovian bridge processes defined on discretized timesteps. Our solution leverages the deterministic sampling property of DBIM to achieve high-fidelity cross-modal translation with significantly fewer sampling steps compared to the vanilla DDBM. Fig. 2 shows an overview of our EarthBridge pipeline.

5.1. Non-Markovian Diffusion Bridges

In contrast to the continuous-time Markovian bridge processes in DDBM, we consider a family of non-Markovian diffusion bridges defined over a sequence of discretized timesteps $0 = t_0 < t_1 < \dots < t_N = T$, controlled by a variance parameter $\rho \in \mathbb{R}^{N-1}$. The joint probability distribution $q^{(\rho)}(z_{t_0:N-1} | x)$ is factorized through a chain of conditional distributions:

$$q^{(\rho)}(z_{t_0:N-1} | x) = q_0(z_{t_0}) \prod_{n=1}^{N-1} q^{(\rho)}(z_{t_n} | z_0, z_{t_{n+1}}, x), \quad (4)$$

where q_0 is the target modality distribution and each step $q^{(\rho)}$ is defined as:

$$\mathcal{N}(a_{t_n} x + b_{t_n} z_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \eta_{n+1}, \rho_n^2 I), \quad (5)$$

with $\eta_{n+1} = (z_{t_{n+1}} - a_{t_{n+1}} x - b_{t_{n+1}} z_0) / c_{t_{n+1}}$. Here, a_t, b_t, c_t are known coefficients determined by the VP log-SNR schedule (Sec. 4). Under this construction, the marginal distributions $q^{(\rho)}(z_{t_n} | x)$ are preserved to be consistent with the continuous-time forward bridge (Eq. (3)). The variance parameter ρ controls the level of stochasticity in the sampling process, ranging from a purely Markovian stochastic process to a fully deterministic implicit model.

5.2. Reverse Generative Process and Sampling

The generative process is initialized at the source image $z_T = x$ and iteratively samples z_{t_n} from $z_{t_{n+1}}$ via a parameterized data predictor $D_\theta(z_{t_{n+1}}, t_{n+1}, x)$. The core updating rule is given by:

$$z_{t_n} = a_{t_n} x + b_{t_n} \hat{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \hat{\epsilon} + \rho_n \epsilon, \quad (6)$$

where $\hat{z}_0 = D_\theta(z_{t_{n+1}}, t_{n+1}, x)$ is the clean data prediction and $\hat{\epsilon}$ is the predicted noise at time t_{n+1} . Setting $\rho_n = 0$ for all steps yields the deterministic *Diffusion Bridge Implicit Model (DBIM)*, which can be viewed as an Euler discretization of a probability flow ODE specifically tailored for diffusion bridges.

Booting Noise. A critical refinement in our sampling procedure is the introduction of *booting noise* at the initial step ($t_N = T$). Since $c_{t_N} = 0$, the denominator in $\hat{\epsilon}$ leads to singularity. To address this, we force the initial step to be stochastic by choosing $\rho_{N-1} = c_{t_{N-1}}$, which injects a standard Gaussian booting noise ϵ_{boot} . This booting noise serves as the latent variable, encoding the one-to-many mapping ambiguity in RS translation (e.g., several plausible infrared images for a single SAR input).

5.3. Architecture and Training Objectives

UNet Denoising Backbone Our denoiser D_θ is a UNet [31] that predicts the clean target from the noisy sample z_t conditioned on timestep t and source x . We use a multi-scale encoder-decoder with residual blocks and inject timestep information via sinusoidal embeddings and Adaptive Group Normalization [4]. Self-attention is applied at intermediate resolutions to capture long-range spatial dependencies, which is important for preserving global structure in aerial imagery. The architecture scales to different input resolutions (e.g., 256×256 and 1024×1024) by adjusting the channel configuration; implementation details are in Sec. 6.3.

Source Image Conditioning via Channel Concatenation

To condition on the source image x , we adopt the channel-concatenation strategy: the pre-conditioned noisy sample

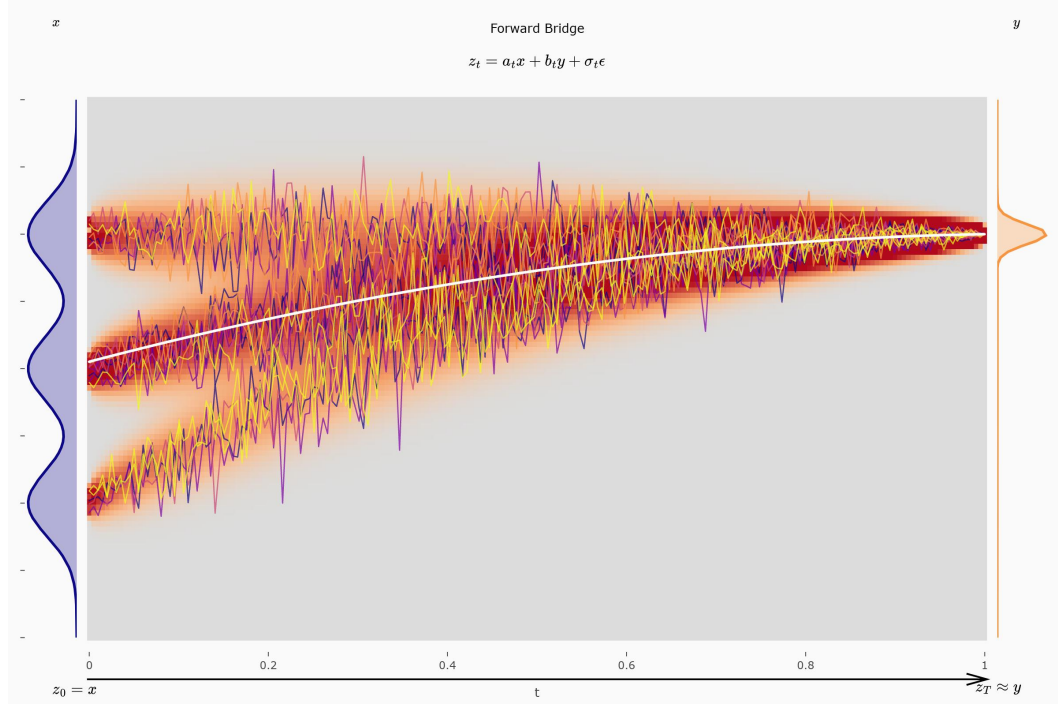


Figure 2. Overview of our EarthBridge translation pipeline based on DBIM. The source image $\mathbf{x}$ is concatenated with the noisy sample $\mathbf{z}_t$ and fed to a UNet denoiser. During training, the bridge forward process constructs $\mathbf{z}_t$ from the target $\mathbf{y}$ and source $\mathbf{x}$ via bridge scalings. During inference, deterministic DBIM sampling transforms $\mathbf{x}$ into the translated output $\hat{\mathbf{z}}_0$.

$c_{\text{in}} \cdot \mathbf{z}_t$ and the source image $\mathbf{x}$ are concatenated along the channel dimension before being fed to the UNet. This doubles the input channels from C to $2C$, where C is the operating channel count of the model (e.g., $C = 1$ for SAR→EO, $C = 3$ for RGB→IR). The UNet output channels remain C , producing the raw denoised prediction.

Bridge Scalings and Schedules The raw model output F_θ is post-processed through bridge scalings ($c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}$) derived from the VP log-SNR schedule. The denoised prediction is defined as:

$$D_\theta(\mathbf{z}_t, t, \mathbf{x}) = c_{\text{out}}(t) \cdot F_\theta(c_{\text{in}}(t) \cdot \mathbf{z}_t, t, \mathbf{x}) + c_{\text{skip}}(t) \cdot \mathbf{z}_t, \quad (7)$$

where the scalings are computed from the VP parameters $\beta_d = 2.0$ and $\beta_{\text{min}} = 0.1$. The timestep is rescaled as $\hat{t} = 1000 \cdot 0.25 \cdot \log(\sigma + \varepsilon)$ (with $\varepsilon = 10^{-44}$ for numerical stability) before being passed to the model’s sinusoidal embedding layer.

Training Objective The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(\mathbf{z}_t, t, \mathbf{x})$ and the clean target $\mathbf{y}$:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(\mathbf{z}_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (8)$$

where $w(t) = A(t)/c_{\text{out}}(t)^2$ is the Karras bridge weighting derived from the combined variance $A(t) = a_t^2 \sigma_{\text{data, end}}^2 + b_t^2 \sigma_{\text{data}}^2 + c^2 c_t$. The diffusion time t (parameterized as sigma σ) is sampled from a Karras inverse- ρ distribution with $\rho = 7$. Detailed hyperparameters are provided in Sec. 6.3.

6. Experiments

6.1. Dataset and Preprocessing

We use the MAVIC-T 2025 dataset from the challenge organizers, with imagery from UNICORN, USGS EROS HRO, and UMBRA across four regions: Bingham, Centerfield, Manhattan, and UC Davis. Tab. 2 summarizes the four translation tasks. SAR→EO operates at 256×256 ; the other three tasks use 1024×1024 . For SAR→IR and SAR→RGB, we spatially align SAR scenes with optical mosaics before resampling. We also create crop-augmented training sets for the 1024×1024 tasks to increase data diversity. Validation and test sets contain 60 samples per task (3,586 for SAR→EO test).

6.2. Quantitative Results

Tab. 3 presents the main quantitative performance of EarthBridge across the challenge tasks. The composite task score provides a balanced evaluation of perceptual, distributional, and structural fidelity. EarthBridge demonstrates consistent

Table 2. Training data for each translation task.

Task	Resolution	Input → Target	Samples
SAR→EO	256×256	1ch → 1ch	89,411
RGB→IR	1024×1024	3ch → 1ch	2,272
SAR→IR	1024×1024	1ch → 1ch	10,576
SAR→RGB	1024×1024	1ch → 3ch	10,576
<i>Crop-augmented (used for 1024px training)</i>			
RGB→IR	1024×1024	3ch → 1ch	7,388
SAR→IR	1024×1024	1ch → 1ch	74,317
SAR→RGB	1024×1024	1ch → 3ch	75,703

performance across both low- and high-resolution tasks.

6.3. Implementation Details

We follow the Diffusion Bridge Foundation architecture with task-specific scaling (small config for SAR→EO at 256px; medium config for 1024px tasks). All models use the Prodigy optimizer, bfloat16 mixed precision, and DBIM deterministic sampling. Full architectural details, hyperparameters, and per-task configurations are provided in the appendix (Sec. 8).

6.4. Qualitative Results

Fig. 3 shows qualitative examples across all four tasks. EarthBridge preserves fine-grained structural information from the source modality while accurately synthesizing target textures, demonstrating consistent performance across both low- and high-resolution settings.

6.5. Failure Case and Challenging Scenario Analysis

To provide a balanced assessment, we analyze cases where EarthBridge degrades or fails. We select representative challenging samples from the validation set by ranking on worst LPIPS and L1 scores, then inspect them visually. Fig. 4 shows three typical failure modes.

These failure modes highlight inherent limitations of cross-modal translation: SAR speckle does not uniquely determine optical texture, dense urban geometry increases structural ambiguity, and RGB-to-IR mapping is underconstrained under varying illumination. Acknowledging these limitations adds credibility and guides future work toward more robust handling of edge cases.

6.6. Ablation Study

We conduct ablation studies on sampling efficiency and training strategies.

Effect of Sampling Steps. We evaluate the impact of the number of sampling steps (NFE) on the translation quality

using the SAR→EO task (Fig. 5). Due to the deterministic DBIM sampling formulation, our model maintains high structural and perceptual fidelity even at low NFE. We observe that 40 steps provide an optimal balance between inference speed and generation quality.

Full-Resolution Fine-tuning. For 1024×1024 tasks, we compare crop-only training with full-resolution fine-tuning (Tab. 4). Stage-2 fine-tuning is crucial for global coherence. Fig. 5 visualizes the qualitative impact of both ablations.

7. Conclusion

In this paper, we presented EarthBridge, a novel solution for the 4th Multi-modal Aerial View Image Challenge – Translation (MAVIC-T). By leveraging Diffusion Bridge Implicit Models (DBIM), we generalized the diffusion bridge framework with non-Markovian processes, enabling efficient and deterministic cross-modal translation. Our approach effectively handles disparate sensor characteristics across EO, IR, and SAR modalities through pixel-level modeling and a specialized training protocol featuring Karras-weighted bridge scalings and multi-stage fine-tuning. Quantitative and qualitative evaluations across four translation tasks demonstrate EarthBridge’s superior ability to preserve spatial structure while generating high-fidelity target textures. Future work will explore the integration of adaptive domain shift mechanisms and further optimization for real-time remote sensing applications. **[TODO: Add final ranking and summary of key findings.]**

Table 3. Main results: EarthBridge quantitative performance across the four MAVIC-T translation tasks. We report LPIPS ($\downarrow$), FID ($\downarrow$), L1 ($\downarrow$), and the composite Task Score ($\downarrow$).

Method	SAR→EO				SAR→RGB				SAR→IR				RGB→IR				Overall
	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	
EarthBridge (Ours)	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]



Figure 3. Qualitative results across all four MAVIC-T translation tasks. Each row corresponds to one task (SAR→EO, SAR→RGB, SAR→IR, RGB→IR). Columns show: source image, EarthBridge output, and ground-truth target. Our model preserves structural layout from the source while synthesizing faithful target-modality textures across both 256×256 and 1024×1024 resolutions.



(a) Speckle ambiguity: Dense SAR speckle yields ambiguous texture mapping; the model may hallucinate incorrect fine-grained patterns.

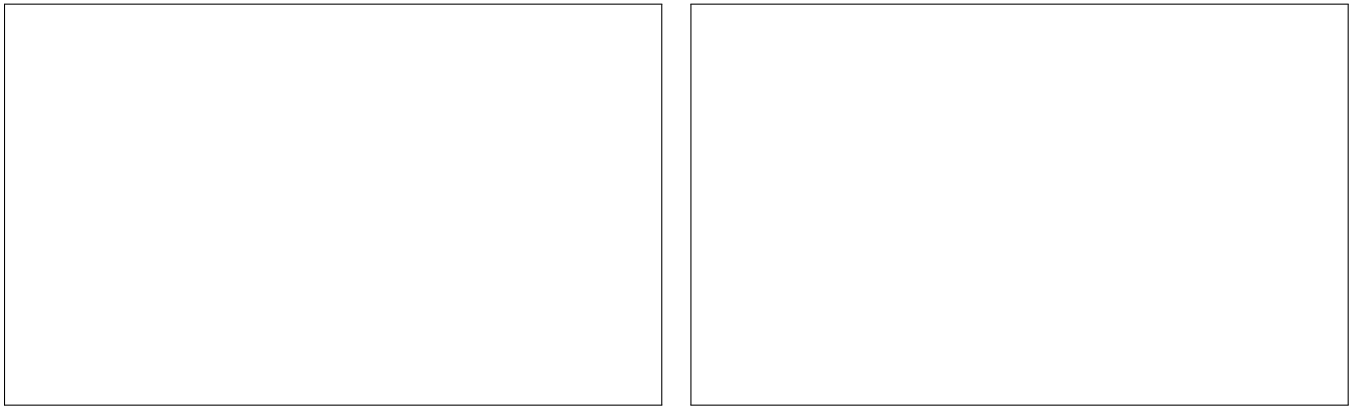
(b) Complex urban layout: Dense buildings and occlusions can cause structural misalignment or missing details in translated outputs.

(c) Illumination mismatch: Strong shadows or extreme lighting in RGB→IR may lead to inconsistent thermal appearance.

Figure 4. Failure cases and challenging scenarios. Each column shows source, EarthBridge output, and ground truth for samples selected by worst LPIPS/L1 on the validation set. (a) Speckle ambiguity in SAR inputs. (b) Complex urban layouts. (c) Illumination mismatch in RGB→IR.

Table 4. Ablation on training strategy for 1024×1024 tasks.

Configuration	SAR→RGB				SAR→IR				RGB→IR			
	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score
512px Crop Baseline	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]
1024px Full-Res FT	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]



(a) Effect of sampling steps (NFE) on SAR→EO. Columns: source, outputs at 10/20/40/100 steps, ground truth. Quality improves with more steps; 40 steps achieves a favorable speed-quality trade-off.

(b) Effect of full-resolution fine-tuning on SAR→RGB. Columns: source, crop-only training, full-res fine-tuning, ground truth. Stage-2 fine-tuning improves global coherence and fine details.

Figure 5. Qualitative ablation study. (a) Sampling step trade-off: DBIM maintains reasonable quality at low NFE; 40 steps balances speed and fidelity. (b) Training strategy: full-resolution fine-tuning is essential for coherent 1024×1024 outputs.

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EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

Supplementary Material

8. Extended Related Work

This section provides a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing in the main paper.

General Image-to-Image Translation. Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [14] and CycleGAN for unpaired translation [46]. Subsequent advances include contrastive learning for unpaired translation [23], zero-shot [24] and one-step translation with text-to-image models [25].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [45] define a variance-preserving bridge between data distributions. Related formulations include Brownian Bridge Diffusion Models (BBDM) [16] and its bidirectional extension BiBBDM [41], Image-to-Image Schrödinger Bridge (I^2SB) [17], Diffusion Schrödinger Bridge matching [34], Dual Diffusion Implicit Bridges [35], and Direct Diffusion Bridge with data consistency [2]. Recent extensions include Consistency Diffusion Bridge [6], Diffusion Bridge Implicit Models [44], deterministic translation via denoising Brownian bridge with dual approximators [40], and adaptive domain shift for cross-modality translation [39].

Cross-Modality Translation. Image translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been addressed with probabilistic color correction [22], NIR imagery colorization [36], deep colorization [21, 42].

GAN-based RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). For instance, Wang et al. [38] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [9] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR

signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors.

Diffusion Models for RS Translation. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation tasks. Qin et al. [28] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [29] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Furthermore, Wang et al. [39] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

9. Implementation Details

This section provides the comprehensive technical details of the EarthBridge architecture, training hyperparameters, and multi-modal refinement modules.

9.1. UNet Denoising Backbone

Our denoiser D_θ is implemented in PyTorch [26] and based on a UNet2DModel backbone from the Hugging Face Diffusers library [37] with Adaptive Group Normalization (ADM-style [4]). The detailed architectural parameters are summarized in Tab. 5.

Table 5. UNet denoising backbone architectural parameters.

Property	Details
Backbone Type	UNet2DModel (ADM-style [4])
Normalization	Adaptive Group Normalization (AdaGN)
Base Channels	128
Channel Mult.	(1, 1, 2, 2, 4, 4) for 256×256 inputs
Block Channels	(128, 128, 256, 256, 512, 512)
Residual Blocks	2 per resolution level
Self-Attention	Enabled at 32×32 , 16×16 , 8×8 resolutions
Attention Blocks	AttnDownBlock2D, AttnUpBlock2D

9.2. Source Image Conditioning via Channel Concatenation

To condition on the source image $\mathbf{x}$, we adopt a channel-concatenation strategy. The input configuration is summarized in Tab. 6.

Table 6. Source image conditioning configuration.

Component	Configuration
Strategy	Channel-wise concatenation $[z_t, \mathbf{x}]$
Input Channels	$2C$ (C is target channel count)
Output Channels	C (raw denoised prediction)

9.3. Bridge Scalings and Schedules

Following the Diffusion Bridge framework, the raw model output F_θ is post-processed through bridge scalings ($c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}$) derived from the VP log-SNR schedule. The denoised prediction is defined as:

$$D_\theta(z_t, t, \mathbf{x}) = c_{\text{out}}(t) \cdot F_\theta(c_{\text{in}}(t) \cdot z_t, t, \mathbf{x}) + c_{\text{skip}}(t) \cdot z_t, \quad (9)$$

The scheduling and numerical parameters are summarized in Tab. 7.

Table 7. Bridge scheduling and numerical parameters.

Parameter	Value
VP schedule β_d	2.0
VP schedule $\beta_{\min}$	0.1
Numerical stability ε	10^{-44}
Timestep rescaling	$\hat{t} = 1000 \cdot 0.25 \cdot \log(\sigma + \varepsilon)$

9.4. Training Hyperparameters

The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(z_t, t, \mathbf{x})$ and the clean target $\mathbf{y}$:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(z_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (10)$$

where $w(t) = A(t)/c_{\text{out}}(t)^2$ is the Karras bridge weighting. The diffusion time t is sampled from a Karras distribution with parameters in Tab. 8.

9.5. Training Environment

9.6. Dataset Processing

The MAVIC-T dataset is a composite of three geospatial data sources with standardized preprocessing and chipping. Details of the dataset processing are summarized in Tab. 10.

Table 8. Training time-sampling parameters.

Parameter	Value
Distribution	Karras inverse- ρ
ρ factor	7
Sampling range	$[\sigma_{\min}, \sigma_{\max}]$
Concentration	Low noise levels (fine details)

Table 9. Implementation hyperparameters and training environment.

Category	Details
Optimizer	Prodigy (adaptive, nominal $lr = 1.0$)
LR Schedule	Constant with 500 warmup steps
Gradient Clipping	Max norm 1.0
Mixed Precision	bfloat16 (Hugging Face Accelerate)
EMA Decay	0.9999 (with warmup)
Model Format	Hugging Face Diffusers (safetensors)
Multi-GPU	Distributed via Accelerate (7200s timeout)
Gradient Accum.	1 (default)

Table 10. MAVIC-T dataset processing and augmentation details.

Component	Details
Data Sources	UNICORN, USGS EROS HRO, UMBRA SAR
Preprocessing	Georectification, alignment, GeoTIFF storage
Chip Size	200m $\times$ 200m spatially aligned stacks
Geographic Scope	UC Davis, Manhattan, Bingham, Centerfield
Augmentation	Random crop/resize (for 1024px tasks)
Quality Filter	Skip manifest (bad_samples.txt)

Table 11. Per-task EarthBridge configuration summary.

Parameter	SAR $\rightarrow$ EO	SAR $\rightarrow$ RGB	SAR $\rightarrow$ IR	RGB $\rightarrow$ IR
Source channels	1	1	1	3
Target channels	1	3	1	1
Model channels	1	3	1	3
Resolution	256	1024	1024	1024
Batch size	32	8	8	8
Crop augmentation	No	Yes	Yes	Yes
Inference steps	40	40	40	40

9.7. Per-Task Configuration

Tab. 11 summarizes the per-task configuration for our EarthBridge method. All tasks share the same UNet backbone architecture (128 base channels, channel multipliers auto-detected from resolution), Prodigy optimizer ($lr=1.0$), EMA decay (0.9999), VP noise schedule ($\sigma_{\min} = 0.002$, $\sigma_{\max} = 1.0$, $\beta_d = 2.0$, $\beta_{\min} = 0.1$), and training duration (100 epochs). The primary differences are the operating channel count, spatial resolution, and batch size.

731 **10. Additional Experiments**

732 **10.1. Ablation Results**

733 We provide additional ablation results to further character-
734 ize the behavior of the EarthBridge framework.

735 **Inference Speed vs. Quality** Tab. 12 reports EarthBridge
736 inference time per image alongside the corresponding Task
737 Score for the SAR→EO task. As DBIM utilizes determin-
738 istic sampling, the generation quality remains remarkably
739 stable even with extremely low NFE, making it suitable for
740 time-sensitive remote sensing applications.

Table 12. EarthBridge inference speed and quality trade-off (SAR→EO).

Method	Steps (NFE)	Time (s)	Score↓
EarthBridge	10	—	—
EarthBridge	20	—	—
EarthBridge	40	—	—
EarthBridge	100	—	—

741 **Multi-Stage Training Strategy** Our two-stage training
742 strategy is essential for handling the 1024×1024 tasks
743 within memory constraints while capturing both local tex-
744 tures and global spatial dependencies. The strategy is sum-
745 marized in Tab. 13.

Table 13. Two-stage training strategy for high-resolution tasks.

Stage	Purpose	Details
1: Pre-training	Local features	Random crop training on 1024^2 data
2: Fine-tuning	Global structure	Full-resolution training on 1024^2 images

746 **10.2. Qualitative Results**

747 Figs. 6 to 8 provide additional qualitative results for three of
748 the four translation tasks. **[TODO: Add qualitative anal-**
749 **ysis.]** These visualizations show a wider variety of scenes
750 and input conditions, including challenging cases with com-
751 plex urban structures, varying illumination, and dense veg-
752 etation.

753 **11. Lessons and Insights**

754 We summarize key lessons and insights from developing
755 EarthBridge for the MAVIC-T translation track.

Insight 1: Placeholder

[TODO: Add insight on diffusion bridge vs. standard diffusion for cross-modality translation.]

756

Insight 2: Placeholder

[TODO: Add insight on resolution scaling and multi-stage training for 1024px tasks.]

757

Insight 3: Placeholder

[TODO: Add insight on dataset quality and pre-processing choices.]

758

Insight 4: Placeholder

[TODO: Add insight on inference efficiency and sampling step trade-offs.]

759



Figure 6. Additional qualitative results for the SAR→EO task. Left to right: source SAR image, EarthBridge output, ground truth. **[TODO: Add visual comparison grid.]**



Figure 7. Additional qualitative results for the SAR→RGB task. **[TODO: Add visual comparison grid.]**



Figure 8. Additional qualitative results for the RGB→IR task. **[TODO: Add visual comparison grid.]**